



Asynchronous RTK positioning enhanced by satellite-based PPP corrections: a case study with PPP-B2b

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Abstract

GNSS network RTK is widely used in autonomous driving and precision mapping due to its high accuracy and coverage. However, NRTK relies on stable communication links. In complex environments such as mountainous or offshore areas, communication interruptions or limited bandwidth can delay the transmission of correction data, thereby degrading positioning accuracy. Asynchronous RTK (ARTK) has attracted increasing attention for its strong tolerance to communication outages. However, with increasing data age, accumulating errors from orbit, clock, ionospheric, and tropospheric delays cannot be fully eliminated by conventional double-difference models, resulting in diminished positioning performance. To address this issue, this paper proposes an asynchronous RTK method enhanced by satellite-based PPP corrections, with PPP-B2b selected as a representative case (B2b-ARTK). It models and compensates the four major error sources. Precise orbit and clock corrections derived from PPP-B2b are applied to mitigate residual errors caused by asynchronous updates. The first-order ionospheric delay is eliminated through a dual-frequency ionosphere-free combination, and the tropospheric delay is corrected using the empirical GPT2w model. Experimental results based on both static and dynamic datasets indicate that, after communication interruptions, the B2b-ARTK method achieves higher positioning accuracy than the ARTK model based on broadcast ephemeris (Brdc-ARTK) and the single-station PPP approach using PPP-B2b corrections (B2b-PPP). In dynamic vehicle experiments, B2b-ARTK maintained a reliable fixed solution for approximately 650 s after communication loss, with a maximum 3D positioning error of 0.085 m, whereas Brdc-ARTK sustained fixed solutions for only 382 s with a maximum error of 0.178 m. These results demonstrate that the B2b-ARTK method effectively mitigates the degradation of RTK performance under unstable communication conditions, ensuring reliable GNSS positioning for applications such as autonomous driving.

Keywords GNSS · Asynchronous RTK (ARTK) · PPP-B2b · Communication interruption · Error compensation

Introduction

Global Navigation Satellite System (GNSS) has become a cornerstone for high-accuracy applications owing to its global coverage and continuous operation. However, traditional standard point positioning (SPP) is limited by factors such as satellite orbit errors, clock errors, ionospheric and

tropospheric delays, making it difficult to achieve centimeter-level accuracy (Guo et al. 2023). To address this, relative positioning methods such as real-time kinematic (RTK) were developed. RTK employs carrier-phase double-differencing between a reference and a rover station to effectively mitigate systematic errors and achieve centimeter-level accuracy, and has been widely used in engineering and agricultural practices. Nevertheless, its performance rapidly deteriorates when the baseline exceeds 10–20 km due to spatially correlated errors such as ionospheric and tropospheric delays (Zhang and Lachapelle 2001; Grejner-Brzezinska et al. 2005; Li et al. 2010). To overcome these baseline limitations, network RTK (NRTK) technology was developed. NRTK establishes a Continuously Operating Reference Station (CORS) network by deploying multiple reference stations within a region to ensure continuous coverage. The

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system collects multi-station observations in real time and applies regional modeling algorithms to estimate and interpolate spatially correlated errors such as ionospheric and tropospheric delays. These interpolated differential corrections are then transmitted to users, enabling a wider service area. Compared to single-baseline RTK, NRTK not only extends the service range but also significantly enhances the accuracy and stability of corrections, thus delivering consistent centimeter-level positioning performance on a regional scale. Owing to its high precision, broad coverage, and adaptability, NRTK has become a cornerstone technology in high-demand applications such as autonomous driving, precision surveying, and engineering monitoring, and it has been widely deployed across China (Hu et al. 2003; Ke et al. 2009). Nevertheless, NRTK performance is highly dependent on stable communication links. In complex environments such as mountainous regions, forests, or dense urban areas, communication between rover and reference stations is often disrupted due to network outages, limited bandwidth, or signal blockage. Such interruptions prevent timely access to correction data and compromise the continuity and robustness of positioning solutions.

To enhance the continuity of GNSS positioning under communication interruptions, asynchronous RTK (ARTK) has been proposed as a solution that balances real-time performance and continuity. In this approach, differential corrections are allowed to have a certain age, and the rover solution is computed using these delayed corrections, which provides tolerance to communication outages. However, existing studies indicate that as correction age increases, orbit, clock, ionospheric, and tropospheric errors gradually accumulate, and residuals that are normally removed by double-differencing (DD) can no longer be completely compensated, thus limiting ARTK accuracy and stability. Shu et al. (2019) systematically analyzed the characteristics of broadcast ephemeris, ionospheric, and clock errors with respect to correction age. They pointed out that the BDS and Galileo systems outperform GPS and GLONASS in terms of error temporal variability, providing a theoretical basis for multi-constellation ARTK error modeling. Under low-latency conditions, research has focused on enhancing high-frequency dynamic positioning while maintaining real-time capability. Zhang et al. (2015) proposed a precise differential ARTK method based on an asynchronous observation model. By introducing backward ephemeris error modeling, the method achieves high-frequency real-time positioning with a delay of several seconds, making it suitable for dynamic applications requiring high-rate positioning. Dong et al. (2020) further integrated time-differenced carrier phase (TDCP) observations to enhance the temporal correlation between adjacent epochs, effectively improving positioning stability under short latency conditions. Paziewski

et al. (2023) integrated high-frequency GNSS observations with accelerometer data and applied RTS smoothing filtering. Their study demonstrated that by reducing the reference station sampling interval to 2 s, the system could recover millimeter-level displacement waveforms in dynamic environments, further extending the performance limits of ARTK for high-frequency dynamic structural monitoring. For high-latency scenarios, studies have sought to mitigate error accumulation through robust models and compensation strategies. Huang et al. (2024) proposed a Kalman filter-based approach for modeling and compensating age-related errors, treating orbit, clock, and atmospheric delays collectively as age errors. By combining truncated Taylor series expansion with sequential filtering, they developed a TDCP model that effectively reduces computational complexity. This method also incorporates a predictor performance monitoring mechanism to evaluate model reliability and provide integrity warnings. However, it requires careful tuning of filter parameters and model structure. Du et al. (2019) proposed an ARTK post-processing strategy based on final precise ephemerides and tropospheric models. This method can maintain centimeter-level positioning accuracy even when base station observations are interrupted for up to 15 min. However, it heavily relies on precise post-processed products, which limits its applicability to real-time scenarios.

In recent years, several satellite-based precise correction services, such as BDS-3 PPP-B2b, Galileo HAS, and QZSS MADOCA/CLAS (Yang et al. 2022; Kawate et al. 2023; Borio et al. 2023), have been made available. These services broadcast precise orbit, clock, and bias corrections directly via satellites, offering the key advantage of independence from terrestrial communication networks, which makes them particularly suitable for ARTK applications under unstable network conditions. However, their periodic update intervals introduce small discontinuities that weaken short-term continuity compared to the smoother polynomial-based broadcast ephemerides. For ARTK, which relies on delayed corrections to maintain accuracy during communication outages, such discontinuities could potentially introduce additional uncertainty, making it essential to examine their impact systematically.

PPP-B2b is selected as a representative case in this study. It is the first globally available and continuously broadcast service, delivering orbit, clock, and differential code bias (DCB) corrections with wide coverage, communication independence, and high accuracy (Xu et al. 2021; Yang et al. 2022). Previous studies have shown that its corrections can achieve centimeter-level orbit and sub-nanosecond clock accuracy, significantly improving Precise Point Positioning (PPP) performance (Jiang et al. 2023; Sun et al. 2023). In this work, a general ARTK framework enhanced

by PPP-B2b corrections is established and its feasibility is systematically evaluated through accuracy analysis and experimental validation.

Within this framework, the method addresses the residual errors inherent in the ARTK model by systematically mitigating four major error sources: satellite orbit errors, satellite clock errors, ionospheric delays, and tropospheric delays. Specifically, (1) orbit and clock parameters obtained from broadcast ephemerides are refined using PPP-B2b corrections, providing more accurate and stable information that reduces their residual impact on positioning; (2) ionosphere-free (IF) combinations are applied to eliminate the first-order ionospheric delay; and (3) leveraging the temporal stability of tropospheric delay (Selmira Garrido et al. 2013), corrections are introduced via the GPT2w model, avoiding the need for real-time estimation. By directly utilizing PPP-B2b broadcasts, the rover independently compensates and models these key error components without relying on NRTK data links or protocol modifications, ensuring strong system compatibility and high flexibility for practical deployment.

The main innovations of this paper include:

- (1) The establishment of a general ARTK framework enhanced by satellite-based PPP corrections, with PPP-B2b selected as a representative case. Within this framework, an orbit and clock error compensation mechanism are introduced to effectively suppress differential residuals under asynchronous conditions;
- (2) A systematic evaluation of orbit and clock accuracy differences between PPP-B2b corrections and broadcast ephemerides, together with quantitative analysis of their impact on ARTK performance. This provides new insights into the continuity and reliability of correction information under different correction ages;
- (3) Comprehensive experimental validation using both static and dynamic datasets, demonstrating that the proposed PPP-B2b-enhanced ARTK method can significantly improve positioning accuracy and availability across diverse application scenarios.

Observation equation of ARTK

Due to frequent interruptions in network communication links, observation data between the rover and reference stations are often temporally asynchronous. Let the observation epochs of the rover and the base station be denoted as t_1 and t_2 , respectively. The undifferenced single-station observation equations for the rover can then be expressed as:

$$\begin{aligned} P_{r,f}^s(t_r) = & \rho_r^s(t_r, t_r^s) + \delta\rho_{orb}^s(t_r^s) \\ & + c \cdot [\delta t_r(t_r) - \delta t^s(t_r^s) - \delta t_{clk}^s(t_r^s)] \\ & + \gamma_f I_{r,1}^s(t_r, t_r^s) + T_r^s(t_r, t_r^s) \\ & + d_{r,f}^{code}(t_r) - d_f^{s,code}(t_r^s) + \varepsilon_{r,f}^s(t_r) \end{aligned} \quad (1)$$

$$\begin{aligned} \Phi_{r,f}^s(t_r) = & \rho_r^s(t_r, t_r^s) + \delta\rho_{orb}^s(t_r^s) \\ & + c \cdot [\delta t_r(t_r) - \delta t^s(t_r^s) - \delta t_{clk}^s(t_r^s)] \\ & + \gamma_f I_{r,1}^s(t_r, t_r^s) + T_r^s(t_r, t_r^s) \\ & + \lambda_f [N_{r,f}^s(t_r) + b_{r,f}^{phase}(t_r^s) - b_f^{s,phase}(t_r^s)] + \eta_{r,f}^s(t_r) \end{aligned} \quad (2)$$

Here, the subscripts r and f denote the receiver and frequency, respectively, while the superscript s represents the satellite. $P_{r,f}^s$ and $\Phi_{r,f}^s$ are the code and carrier-phase observations (in meters), respectively. t_r denotes the signal reception time at the receiver, with $t_r = t_1 + \delta t_r$, and t_r^s denotes the signal transmission time from the satellite, with $t_r^s = t_r - \rho_r^s/c$. ρ_r^s is the geometric distance between the satellite and receiver, and c is the speed of light in vacuum. $\delta\rho_{orb}^s$ represents the satellite orbit error; δt_r and δt^s are the receiver and satellite clock biases, respectively, and δt_{clk}^s denotes the satellite clock bias error. γ_f is the ionospheric delay mapping factor of frequency f relative to frequency 1, and I_r^s is the ionospheric delay at frequency 1. T_r^s represents the tropospheric delay. λ_f is the carrier wavelength of frequency f , and $N_{r,f}^s$ is the integer carrier-phase ambiguity. $d_{r,f}^{code}$ and $d_f^{s,code}$ denote the code hardware delays at the receiver and satellite ends for frequency f , respectively; $b_{r,f}^{phase}$ and $b_f^{s,phase}$ denote the corresponding carrier-phase hardware delays. $\varepsilon_{r,f}^s$ and $\eta_{r,f}^s$ represent the measurement noise of the code and carrier-phase observations, including multipath effects.

Error analysis and mitigation methods for ARTK

As shown in Fig. 1, let the reference station be denoted as b and the rover station as r . Assume that the satellite is observed at position p at the base station epoch and at position p' at the rover epoch. For this satellite, the single-differenced (SD) carrier-phase observation equation between the rover and the base station can be expressed as:

$$\begin{aligned} \Delta\Phi_{rb,f}^s(t_r, t_b) = & \Delta\rho_{rb}^s(t_r, t_b, t_r^s, t_b^s) + \Delta\delta\rho_{orb}^s(t_r^s, t_b^s) \\ & + c \cdot [\Delta\delta t_{rb}(t_r, t_b) - \Delta\delta t^s(t_r^s, t_b^s) \\ & + \Delta\delta t_{clk}^s(t_r^s, t_b^s)] \gamma_f \Delta I_{rb,1}^s(t_r, t_b, t_r^s, t_b^s) \\ & + \Delta T_{rb}^s(t_r, t_b, t_r^s, t_b^s) + \lambda_f \Delta N_{rb,f}^s(t_r, t_b) \\ & + \Delta\eta_{r,f}^s(t_r, t_b) \end{aligned} \quad (3)$$

It is worth noting that, in conventional synchronous RTK (SRTK) models, spatially correlated errors such as satellite

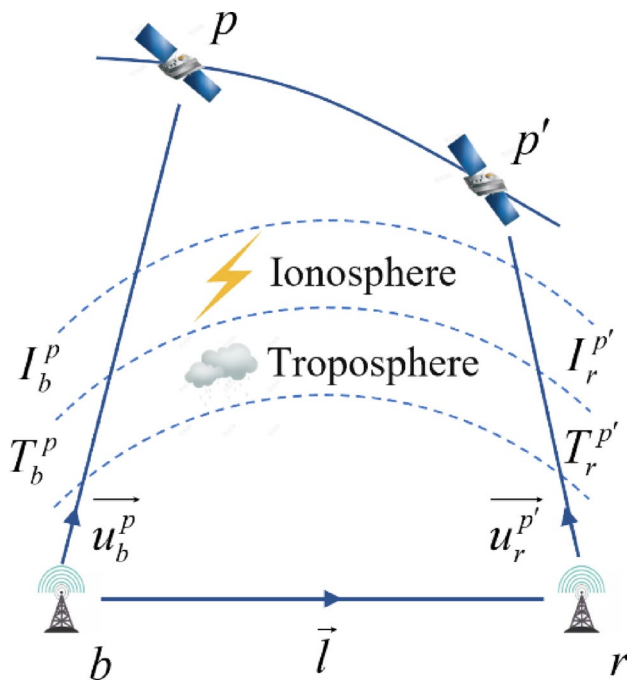


Fig. 1 Asynchronous inter-station single-difference schematic

orbit errors, satellite clock bias errors, ionospheric delays, and tropospheric delays can be effectively mitigated or eliminated through inter-station differencing. Under asynchronous conditions, however, the temporal mismatch between the observation epochs of the reference and rover stations introduces significant time dependency into these error components. As a result, residual errors remain after differencing and become a major source of degraded ARTK positioning accuracy. In contrast, hardware-related biases such as receiver and satellite code delays as well as carrier phase delays tend to exhibit strong temporal stability. Therefore, even with non-simultaneous observations, these

biases can still be largely eliminated by inter-station differencing, exerting only a minor influence on ARTK positioning accuracy (Guo et al. 2015; Zeng et al. 2023).

To mitigate the degradation caused by such residual errors, this study develops a high-precision ARTK framework enhanced by satellite-based PPP corrections, with BDS-3 PPP-B2b selected as a representative implementation (hereafter referred to as B2b-ARTK). The method systematically models and compensates for four primary error sources: orbit errors, clock bias errors, ionospheric delays, and tropospheric delays, thereby improving both the accuracy and stability of ARTK positioning. The overall algorithm framework is shown in Fig. 2.

The specific analysis and correction methods for the four types of errors are described as follows.

(1) Satellite Orbit Error

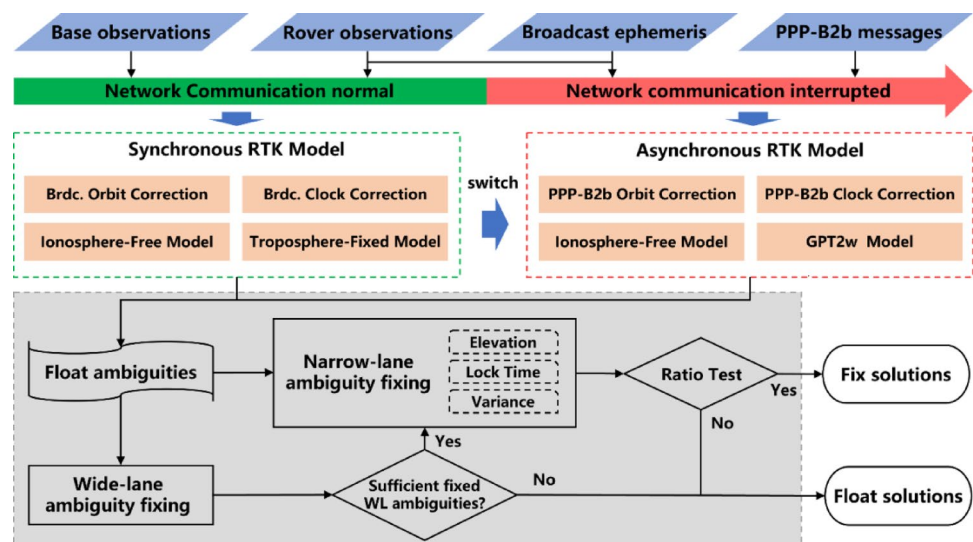
The residual component of the satellite orbit error in the SD observations between stations can be expressed as:

$$\Delta\delta\rho_{orb}^{pp'}(t_r^s, t_b^s) = u_r^{p'}(t_r^s) \cdot \delta_{orb}^{p'}(t_r^s) - u_b^p(t_b^s) \cdot \delta_{orb}^p(t_b^s) \quad (4)$$

To further analyze the error structure of this term, we transform and decompose it, resulting in the following expression:

$$\begin{aligned} \Delta\delta\rho_{orb}^{pp'}(t_r^s, t_b^s) &= \underbrace{\left[u_r^{p'}(t_r^s) - u_b^p(t_b^s) \right] \cdot \delta_{orb}^{p'}(t_r^s)}_{\text{(a) direction-induced term}} \\ &+ \underbrace{u_b^p(t_b^s) \cdot \left[\delta_{orb}^{p'}(t_r^s) - \delta_{orb}^p(t_b^s) \right]}_{\text{(b) time-induced term}} \end{aligned} \quad (5)$$

Fig. 2 Overall framework of the B2b-ARTK model



Here, $\delta_{orb}^p(t_b^s)$ represents the orbit error of satellite p at epoch t_b^s . Equation (5) shows that orbital residuals in ARTK originate from two types of asynchrony:

- (a) Directional asynchrony: Since the satellite positions at epochs t_r^s and t_b^s differ, the unit line-of-sight vectors $u_{r'}^{p'}$ and u_b^p are not identical, leading to unequal projections of orbit errors. This component grows with baseline length, satellite velocity, and the observation time interval between the base and rover stations, and it still causes some impact in SRTK, though the effect is relatively limited.
- (b) Temporal asynchrony: As orbit errors vary over time, the difference between t_r^s and t_b^s along the same direction generates additional residuals. This effect is negligible in SRTK but becomes significant in ARTK.

Overall, ARTK orbit residuals are determined by the time offset, satellite velocity, baseline length, and ephemeris accuracy. This error becomes especially significant when satellite orbits change rapidly or when low-precision ephemerides are used (e.g., broadcast ephemerides). In such cases, the residuals can greatly exceed those in SRTK and must be explicitly modeled or corrected.

To suppress these effects, high-accuracy orbit corrections are indispensable. The BDS-3 system provides high-accuracy real-time orbit correction data through its PPP-B2b service, which enables dynamic compensation for the orbital errors present in broadcast ephemerides and achieves better accuracy than conventional broadcast products. The orbit correction information broadcast by the PPP-B2b signal includes radial (R), along-track (A), and cross-track (C) components, which can be used to correct the satellite position derived from broadcast ephemerides as follows (CSNO 2020).

$$\begin{aligned} \mathbf{X}_{B2b} &= \mathbf{X}_{Brdc} - [\mathbf{e}_{\text{radial}} \quad \mathbf{e}_{\text{along}} \quad \mathbf{e}_{\text{cross}}] \cdot \delta \mathbf{O} \\ \mathbf{e}_{\text{radial}} &= \frac{\vec{r}_s}{|\vec{r}_s|}, \quad \mathbf{e}_{\text{cross}} = \frac{\vec{r}_s \times \vec{v}_s}{|\vec{r}_s \times \vec{v}_s|}, \quad \mathbf{e}_{\text{along}} = \mathbf{e}_{\text{cross}} \times \mathbf{e}_{\text{radial}} \end{aligned} \quad (6)$$

In Eq. (6), $\vec{r}_s$ and $\vec{v}_s$ denote the satellite position and velocity vectors, respectively; $\delta \mathbf{O}$ is the PPP-B2b orbit correction vector; $\mathbf{X}_{Brdc}$ represents the satellite position computed from broadcast ephemeris (LNAV for GPS and CNAV1 for BDS); and $\mathbf{X}_{B2b}$ is the corrected high-precision satellite position.

(2) Satellite Clock Bias Error

In ARTK positioning, time offsets between the observation epochs of the base and rover stations prevent complete elimination of satellite clock biases by inter-station differencing.

As a result, residual clock bias errors are introduced. Their magnitude is determined primarily by the short-term stability of onboard atomic clocks and the variation rate of satellite clock biases (Griggs et al. 2014, 2015).

Taking current mainstream GNSS systems as examples, the average standard deviation of satellite clock bias in the GPS system using LNAV broadcast ephemerides is approximately 0.25 ns, while that of the BDS-3 system using CNAV1 broadcast ephemerides is about 0.6 ns. Under a data latency of 30 s, these could respectively introduce residual clock bias errors of approximately 0.075 m and 0.18 m (Liu et al. 2022; Peng et al. 2024). Although such errors are on the decimeter level, they constitute a non-negligible error source in RTK systems that require centimeter-level positioning accuracy. This is especially critical under conditions of large data latency or high environmental dynamics, where the impact on positioning accuracy becomes more pronounced. Therefore, the performance evaluation of ARTK systems should fully account for the clock stability characteristics of each GNSS constellation and incorporate prediction or correction models to mitigate the influence of residual errors on positioning accuracy.

Compared with broadcast models, which use low-order polynomials and offer smooth but less accurate clocks, PPP-B2b provides real-time clock corrections with significantly higher accuracy, improving both the precision and stability of ARTK. The corrected clock bias is obtained as:

$$\delta t_{B2b}^s = \delta t_{Brdc}^s - \Delta t_{B2b} \quad (7)$$

where Δt_{B2b} denotes the clock corrections provided by PPP-B2b (unit: meters), δt_{Brdc}^s represents the satellite clock bias calculated from broadcast ephemerides, and δt_{B2b}^s is the precise clock bias after PPP-B2b correction.

Although PPP-B2b corrections substantially improve orbit and clock accuracy, their periodic update intervals (6 s for clock and 48 s for orbit) introduce small discontinuities. These may cause short-term fluctuations in low-latency ARTK solutions. Nevertheless, the long-term benefits in accuracy and stability significantly outweigh these minor drawbacks compared with broadcast ephemerides.

(3) Ionospheric Delay

The residual ionospheric delay term in the SD between stations can be expressed as:

$$\begin{aligned}
\Delta I_{rb}^{pp'}(t_r, t_b, t_r^s, t_b^s) &= I_r^{p'}(t_r, t_r^s) - I_b^p(t_b, t_b^s) \\
&= \underbrace{\left[I_r^{p'}(t_b, t_b^s) - I_b^p(t_b, t_b^s) \right]}_{\text{(a) direction - induced term}} \\
&\quad + \underbrace{\left[I_r^{p'}(t_r, t_r^s) - I_r^{p'}(t_b, t_b^s) \right]}_{\text{(b) time - induced term}}
\end{aligned} \quad (8)$$

In ARTK, ionospheric residuals after inter-station differencing arise from both spatial inconsistencies between stations and temporal variations during the observation interval. While the temporal variation rate is usually stable over short timescales (Zhu et al. 2023), residuals can grow under strong ionospheric activity or larger latency. To mitigate their impact, two approaches are commonly used: (1) ionosphere-free (IF) linear combinations to remove the first-order ionospheric delay, or (2) prediction models based on delay variation rates. In this study, the IF combination is employed, which reliably suppresses ionospheric residuals introduced by asynchronous observations.

(4) Tropospheric delay

The residual tropospheric delay in the SD observation between stations can be expressed as:

$$\begin{aligned}
\Delta T_{rb}^{pp'}(t_r, t_b, t_r^s, t_b^s) &= T_r^{p'}(t_r, t_r^s) - T_b^p(t_b, t_b^s) \\
&= \underbrace{\left[T_r^{p'}(t_b, t_b^s) - T_b^p(t_b, t_b^s) \right]}_{\text{(a) direction - induced term}} \\
&\quad + \underbrace{\left[T_r^{p'}(t_r, t_r^s) - T_r^{p'}(t_b, t_b^s) \right]}_{\text{(b) time - induced term}}
\end{aligned} \quad (9)$$

Similar to the ionosphere, tropospheric residuals in ARTK stem from spatial path differences and temporal variability, with the wet component being more sensitive to rapid atmospheric changes. However, short-term tropospheric variations are generally small and stable (Pei et al. 2024). Considering both efficiency and accuracy, this study adopts the GPT2w model, which corrects ~95% of the dry delay

with millimeter-level accuracy (Liu et al. 2017). Although the wet delay remains more variable, its magnitude is comparatively minor (typically ~5% of the total delay), making GPT2w an effective solution for suppressing tropospheric residuals under asynchronous conditions without adding computational burden. Nevertheless, it should be noted that GPT2w assumes relatively stable meteorological conditions, and its performance may degrade during extreme weather events, a limitation that needs to be considered in practical applications.

In summary, this study systematically analyzes the residual mechanisms in ARTK by examining the temporal and spatial characteristics of four main error sources: orbit errors, clock biases, ionospheric delays, and tropospheric delays. The core challenge of ARTK lies in the epoch mismatch between base and rover observations, which breaks the assumption of cancelable “common errors” in synchronous models and introduces significant residuals after differencing. To mitigate this, targeted strategies are employed for each error source, thereby alleviating accuracy degradation under asynchronous conditions. Table 1 summarizes the error characteristics, residual causes, and corresponding mitigation methods.

Satellite orbit and clock bias accuracy evaluation method

To assess the impact of PPP-B2b products on ARTK performance, this study evaluates the orbit and clock accuracy of broadcast ephemerides before and after PPP-B2b correction, using CODE precise products as references. The CODE solutions provide orbits at 5-min intervals and clocks at 30-s intervals.

(1) Satellite Orbit Accuracy Assessment

When evaluating orbit accuracy, two consistency issues must be addressed. First, CODE precise products are referenced to the satellite center of mass, while broadcast ephemerides and PPP-B2b-corrected positions are referred to the antenna phase center. To ensure consistency, all positions are converted to the center of mass, applying only the phase center offset (PCO) correction since phase center variation (PCV) is negligible (Chen et al. 2018). Second, CODE products are expressed in the IGS reference frame, whereas broadcast ephemerides of different constellations adopt their own systems (e.g., WGS84 for GPS, CGCS2000 for BDS). A seven-parameter transformation is applied to unify all products to the IGS frame (Hou and Wang 2016). Orbit errors are then evaluated in the radial-along-cross (RAC) directions as:

Table 1 Summary of residual error causes, and mitigation strategies for major error sources in ARTK

Error Type	Causes	Mitigation strategies
Orbit error	Time Asynchrony and Directional Difference	PPP-B2b Correction
Clock bias error	Time Asynchrony	PPP-B2b Correction
Ionospheric delay	Time Asynchrony and Spatial Path Difference	IF Combination
Tropospheric delay	Time Asynchrony and Spatial Path Difference	GPT2w Model

$$[\delta R \ \delta A \ \delta C]^T = \mathbf{R}_2 \cdot [(\mathbf{X}_B + \mathbf{R}_1 \cdot \mathbf{V}_{PCO_{SYS}}) - \mathbf{X}_{CODE}] \quad (10)$$

Here, $\mathbf{X}_{CODE}$ represents the satellite position calculated from the CODE precise orbit, and $\mathbf{X}_B$ represents the satellite position from the broadcast ephemeris or corrected by PPP-B2b. $\mathbf{R}_1$ denotes the satellite attitude matrix. $\mathbf{R}_2$ is the transformation matrix from the Earth-Centered Earth-Fixed (ECEF) coordinate system to the RAC frame of the satellite.

In Eq. (10), $\mathbf{V}_{PCO_{SYS}}$ denotes the satellite antenna PCO correction vector. Since the GPS LNAV ephemeris uses the ionosphere-free combination of L1 and L2, while the BDS-3 CNAV1 ephemeris refers to B3I, the $\mathbf{V}_{PCO}$ for GPS and BDS-3 are computed as:

$$\mathbf{V}_{PCOG} = \frac{f_1^2}{f_1^2 - f_2^2} \mathbf{V}_{L1} - \frac{f_2^2}{f_1^2 - f_2^2} \mathbf{V}_{L2}, \mathbf{V}_{PCOC} = \mathbf{V}_{B3} \quad (11)$$

Here, $f_1 = 1575.42$ MHz and $f_2 = 1227.60$ MHz denote the carrier frequencies of GPS signals L1 and L2, respectively; $\mathbf{V}_{L1}$ and $\mathbf{V}_{L2}$ represent the antenna PCO correction vectors at the L1 and L2 frequencies; $\mathbf{V}_{B3}$ represents the antenna PCO correction vector at the BDS B3 frequency.

(2) Satellite Clock Bias Accuracy Assessment

In real-time processing, satellite clock biases are typically estimated under known orbits and station coordinates using an ionosphere-free (IF) combination (Yao et al. 2017). These clock bias parameters mainly consist of the following three components:

1. *Reference Clock Bias Term t_0* : Because receiver and satellite clocks are linearly correlated, fixing a reference clock is necessary to avoid rank deficiency. This introduces a common offset shared by all satellites at each epoch. Different products may adopt different reference clocks, leading to systematic offsets between them.
2. *Relative Clock Bias Term Δt^s* : Carrier phase dominates clock estimation due to its high precision, but ambiguities are also correlated with the satellite clock. Applying epoch differencing (ED) removes ambiguities and yields relative clock biases between consecutive epochs.
3. *Initial Clock Bias Term t_0^s* : After ED provides relative values, the first-epoch bias is initialized from pseudorange observations. The accuracy of this initial offset depends on pseudorange quality.

Therefore, the satellite clock bias can be expressed by the following equation:

$$\delta t^s = t_0 + t_0^s + \Delta t^s \quad (12)$$

When evaluating broadcast and PPP-B2b clocks against CODE precise products, two alignment issues must be addressed:

- (1) Frequency reference alignment. PPP-B2b clocks follow the broadcast reference, which may differ from the precise product. For instance, BDS-3 broadcast clocks use B3I, whereas CODE adopts a B1I/B3I combination; GPS clocks are referenced to L1/L2 IF. Therefore, only BDS-3 requires a differential code bias (DCB) correction, applied as:

$$\delta t_{\text{corr}}^s = \delta t_B^s + \frac{f_{1C}^2}{f_{1C}^2 - f_{3C}^2} \cdot DCB_{B3I}^s \quad (13)$$

- (2) In the Eq. (13), DCB_{B3I}^s denotes the DCB between the B1I and B3I frequencies for BDS-3 satellites; $f_{1C} = 1575.42$ MHz and $f_{3C} = 1268.52$ MHz are the carrier frequencies of the B1I and B3I signals in the BDS-3 system, respectively. δt_B^s represents the satellite clock bias from the broadcast ephemeris or corrected by PPP-B2b, and δt_{corr}^s denotes the satellite clock bias after DCB correction.
- (3) Time reference alignment. Broadcast and precise clocks use different reference clocks, producing a constant system offset across all satellites and epochs, which SD cannot remove. This offset is corrected epoch by epoch using the mean difference over tracked satellites:

$$\nabla \Delta \delta t^s = \Delta \delta t^s - \frac{\sum_{s=1}^n \Delta \delta t^s}{n} \quad (14)$$

$$\Delta \delta t^s = \delta t_{\text{corr}}^s - \delta t_{\text{code}}^s \quad (15)$$

Here, δt_{code}^s denotes the satellite precise clock bias from CODE, n is the number of observed satellites at the current epoch, and $\nabla \Delta \delta t^s$ represents the final clock bias accuracy.

It is worth noting that when the number or type of observed satellites changes between epochs, the mean correction method may cause abrupt changes in the clock bias evaluation results. To address this issue, this study further introduces a continuity constraint for smoothing, as shown in the following equation:

$$\sum_{s=1}^m \nabla \Delta \delta t^s(k) = \sum_{s=1}^m \nabla \Delta \delta t^s(k-1) \quad (16)$$

Here, m represents the number of common satellites observed at epochs k and $k-1$.

Experiment and analysis

The experimental section of this study consists of four parts to comprehensively validate the performance and applicability of the proposed models. First, three positioning strategies are defined: the broadcast-based ARTK model (Brdc-ARTK), the PPP-B2b-enhanced ARTK model (B2b-ARTK), and the single-station PPP model using PPP-B2b corrections (B2b-PPP). Second, the orbit and clock accuracies of broadcast ephemerides and PPP-B2b corrections for both BDS-3 and GPS are assessed. Using precise products as references, residual errors are statistically analyzed under different correction ages, providing the basis for subsequent positioning comparisons. Third, positioning performance under various fixed correction ages is evaluated by comparing Brdc-ARTK and B2b-ARTK, focusing on differences in fixed rate and fixed solution accuracy. Finally, static short-baseline data and vehicle-borne dynamic data are used to simulate network outages, where the three models are tested for positioning accuracy and solution continuity. Particular emphasis is placed on the ability of Brdc-ARTK and B2b-ARTK to maintain continuous fixed solutions, thereby verifying the practicality and advantages of the proposed B2b-ARTK framework in asynchronous environments.

Model formulation and experimental data

(1) System Model Establishment

To ensure the robustness and reliability of the validation, this study conducts comparative experiments involving three positioning models: B2b-ARTK, Brdc-ARTK, and B2b-PPP. The basic processing configurations for each model are listed in Table 2. Model-based correction strategies are applied for tidal effects and antenna phase wind-up errors, consistent with the standard PPP configuration (Teunissen et al. 2013).

(2) Data Collection

To comprehensively evaluate the positioning performance of the proposed B2b-ARTK model, one static dataset and one kinematic vehicle-borne dataset are used for experimental evaluation. The basic information of the two datasets is summarized in Table 3.

The static data were collected from the HKWS-HKSS baseline in Hong Kong, with observation data accessible in real time via NTRIP (Hostname: ntrip.geodetic.gov.hk; Port: 2101). The dynamic data were collected during a field campaign conducted by the authors. A nearby static reference station was used as the base station, and the reference

Table 2 Processing strategies of the three positioning models

Item	Brdc-ARTK	B2b-ARTK	B2b-PPP
Observations	Rover and Base Station	Rover and Base Station	Rover Only
Signal selection	GPS (L1, L2) BDS (B1, B3)	GPS (L1, L2) BDS (B1, B3)	GPS (L1, L2) BDS (B1, B3)
Orbit and clock	Broadcast	Broad- cast+PPP-B2b	Broad- cast+PPP-B2b
Tidal effects	Solid earth and ocean tides model correction	Solid earth and ocean tides model correction	Solid earth and ocean tides model correction
Phase windup	Correction	Correction	Correction
Tropospheric delays	GPT2w	GPT2w	Correct ZTD and estimate ZWD
Ionospheric delays	IF combination	IF combination	IF combination
Elevation mask	10°	10°	10°

Table 3 Basic information of the two datasets

Item	Data Type	
	Static	Vehicle
Receiver type	Leica GR50	Sino K803
Receiver antenna type	LEIAR25.R4 LEIT	AT312 (CHCNAV)
Signal selection	Double-frequency	Double-frequency
Satellite system	GPS/BDS-3	GPS/BDS-3
Sampling rate	1 s	1 s
Baseline length	5.6 km	5.5–7.6 km



Fig. 3 Vehicle trajectory during the dynamic test (green line) and location of the base station (red star)

trajectory was provided by the NovAtel SPAN-ISA-100C integrated navigation system and processed using Inertial Explorer (IE) software. Figure 3 shows the vehicle trajectory during the dynamic experiment, demonstrating that the testing environment was relatively unobstructed.

To ensure consistency and fairness of the comparison results, both static and dynamic datasets were processed using only satellites that correctly matched the PPP-B2b correction information and met the positioning conditions.

As shown in Fig. 4, the number of available satellites in the static dataset remained relatively stable, with an average of 14.4 satellites visible and the position dilution of precision (PDOP) consistently below 2.0. In contrast, the dynamic dataset is more affected by environmental variations, with correspondingly higher PDOP values, and the average number of visible satellites was 12.5.

Broadcast versus PPP-B2b orbit and clock accuracy assessment

To assess the impact of PPP-B2b products on enhancing orbit and clock accuracy for ARTK positioning, it is essential to evaluate the orbit and clock accuracy of the broadcast ephemeris and PPP-B2b ephemeris prior to the positioning experiments. Considering that radial orbit errors have the most significant influence on positioning accuracy (Montenbruck et al. 2018), this study focuses on evaluating the radial orbit errors of both broadcast and PPP-B2b ephemerides. Since ARTK emphasizes error stability rather than absolute magnitude, the standard deviation (STD) of radial orbit errors and clock biases is used as the primary metric. Given that error characteristics may vary across different satellite visibility arcs, the STD is first computed for each arc segment of each satellite, and the average of these values is then taken as the overall STD for that satellite. The computation process is described as follows:

$$\text{STD}^s = \frac{1}{M} \sum_{j=1}^M \sqrt{\frac{1}{N_j} \sum_{i=1}^{N_j} (x_i^{(j)} - \bar{x}^{(j)})^2} \quad (17)$$

where M denotes the number of arc segments for the satellite, N_j is the number of epochs in arc segment j ; $x_i^{(j)}$ represents the satellite error at epoch i in arc segment j ; $\bar{x}^{(j)}$ is

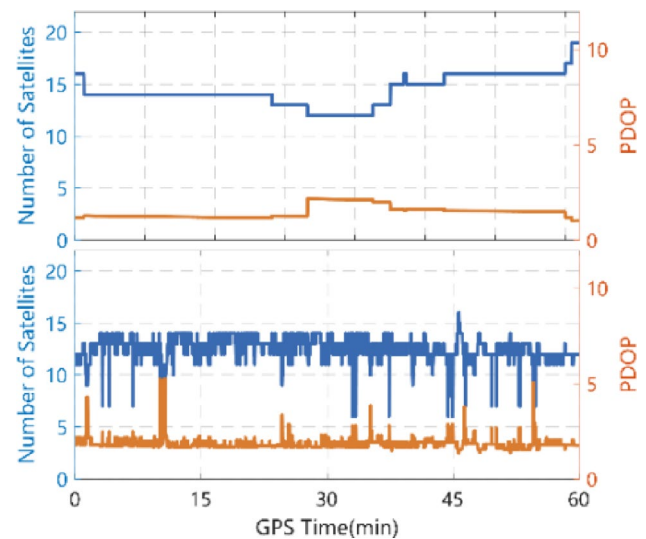


Fig. 4 Available satellites and PDOP in static (top) and dynamic (bottom) experiments

the mean error in arc segment j ; and STD^s denotes the final standard deviation of the satellite error.

As shown in Fig. 5, the STD of the satellite radial orbit errors for the GPS system is reduced from 0.276 to 0.054 m after applying the PPP-B2b corrections, indicating a significant improvement in both orbit accuracy and stability. In contrast, the BDS-3 CNAV1 ephemeris already shows better orbit accuracy and stability than the GPS LNAV ephemeris, which is consistent with previous studies attributing this advantage to the inter-satellite link (ISL) observations employed in the BDS-3 system that enhance orbit determination precision (Li et al. 2022; Xie et al. 2025). After PPP-B2b correction, its standard deviation is further reduced from 0.099 to 0.048 m. This confirms that PPP-B2b corrections are effective not only for GPS but also for further enhancing the higher-quality BDS-3 broadcast ephemeris.

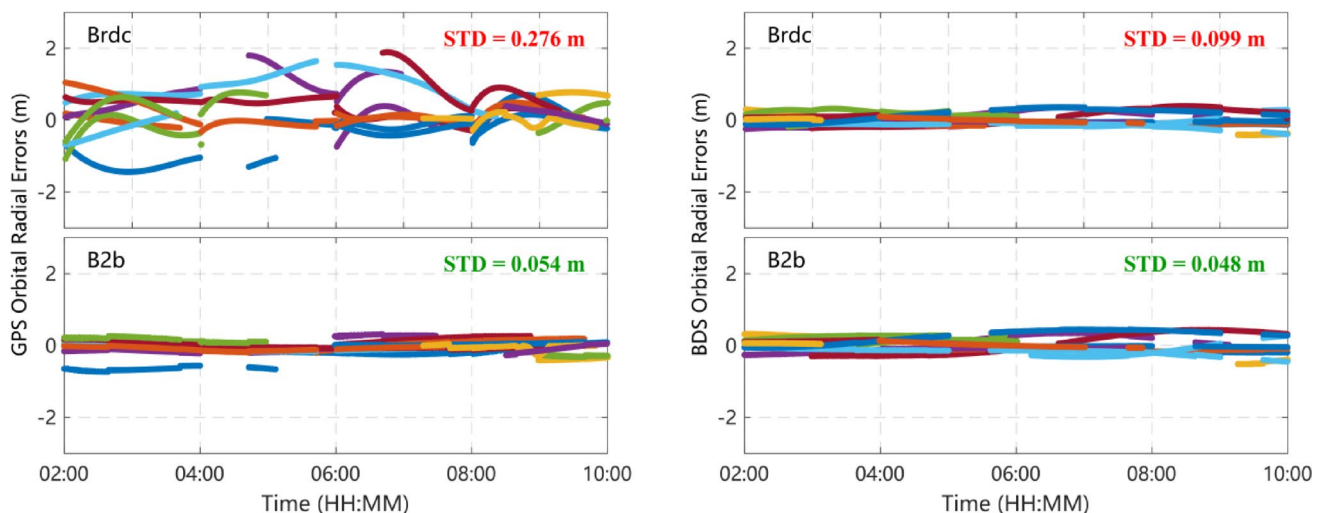


Fig. 5 Comparison of radial orbit errors between broadcast and PPP-B2b ephemeris for GPS (left) and BDS-3 (right)

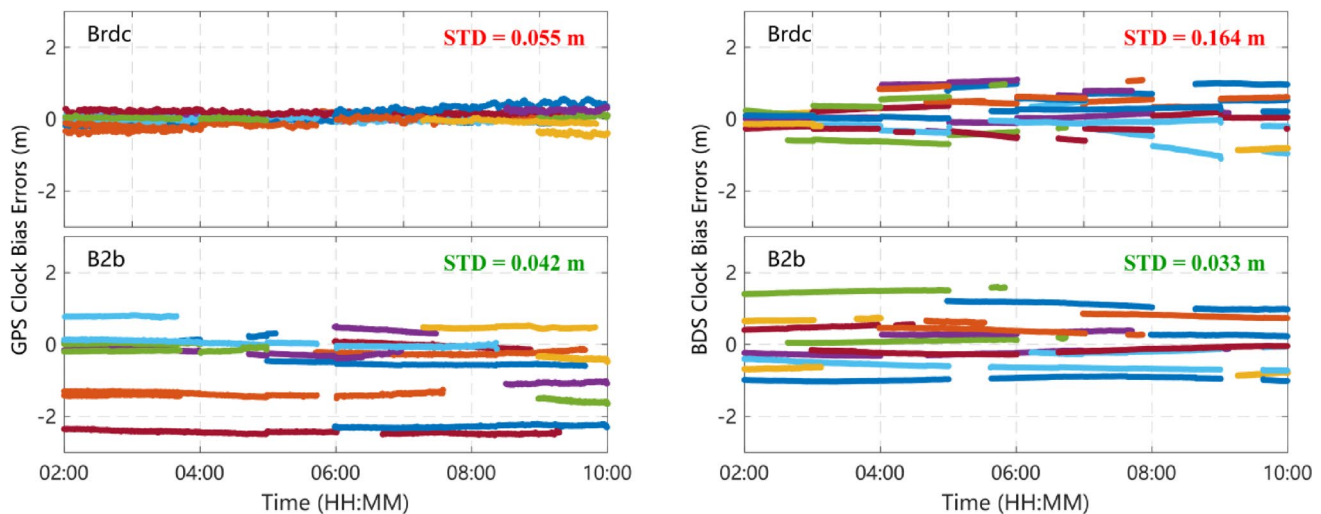


Fig. 6 Comparison of clock bias errors between broadcast and PPP-B2b ephemeris for GPS (left) and BDS-3 (right)

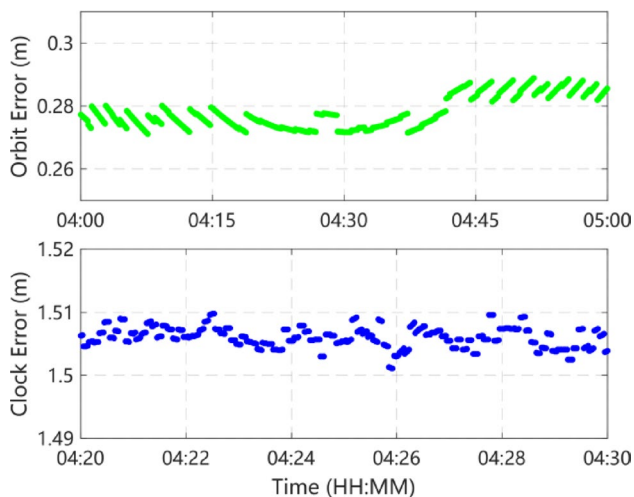


Fig. 7 Local residuals of radial orbit (top) and clock bias (bottom) errors for satellite C30 after applying PPP-B2b corrections

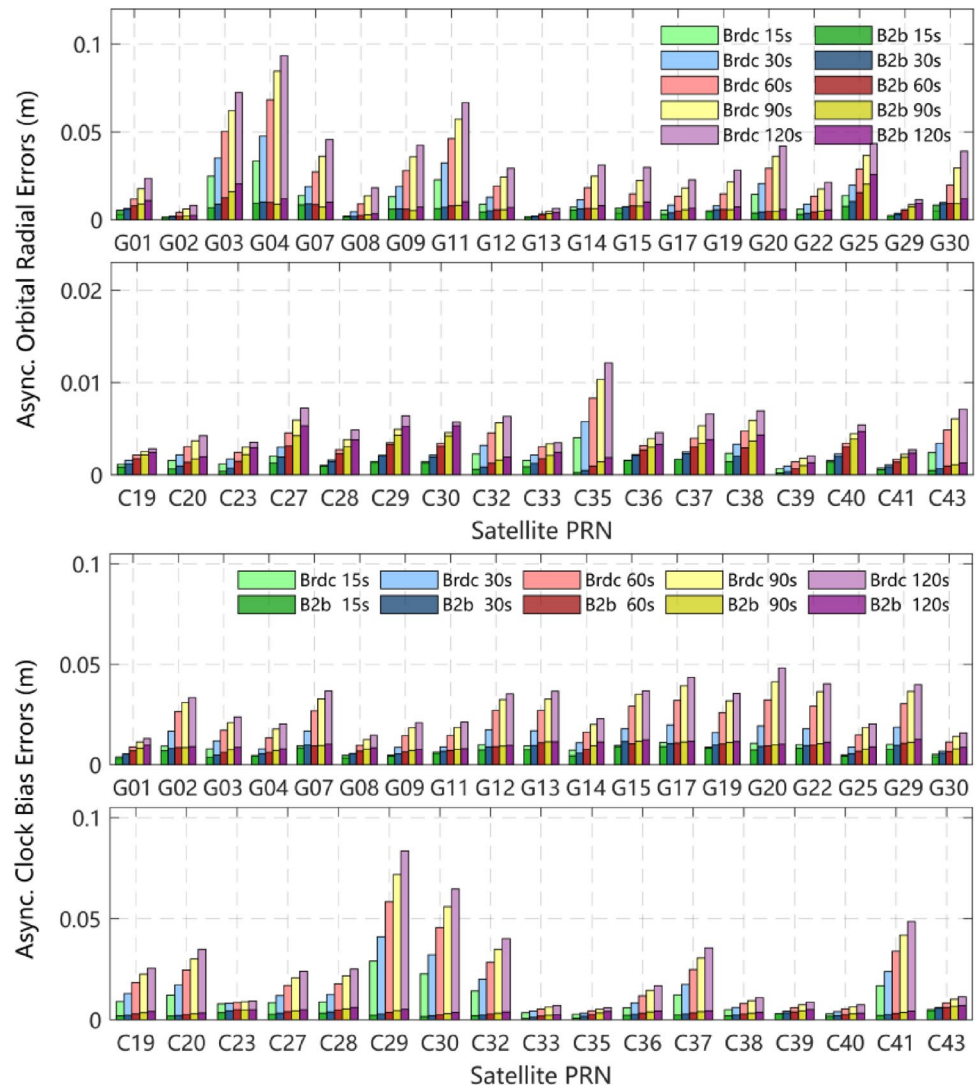
As shown in Fig. 6, after applying PPP-B2b corrections, a constant bias is observed in the clock biases of both GPS and BDS-3 systems. This constant bias originates from the initialization of satellite clock estimation, where pseudorange observations are used to provide the absolute reference at the beginning of each arc, thereby introducing satellite-specific offsets that remain stable within an arc (Sun et al. 2023). These biases are carried into both pseudorange and carrier-phase equations. While they do not affect the final float PPP accuracy since they are eventually absorbed by the ambiguity parameter, they can slow down the ambiguity resolution process and thus reduce PPP convergence speed. However, since ARTK positioning performance is primarily affected by the stability of clock biases, this constant bias has a limited impact on the overall positioning accuracy. Additionally, the GPS LNAV ephemeris demonstrates better clock stability compared to BDS-3. After the application of

PPP-B2b corrections, the standard deviation of GPS clock bias errors is reduced from 0.055 to 0.042 m, while that of BDS-3 is significantly reduced from 0.164 to 0.033 m.

To further highlight the short-term behavior of PPP-B2b corrections, a local time series of orbit and clock errors for a representative BDS-3 satellite (C30) is shown in Fig. 7. The results clearly reveal small periodic jumps: orbit corrections exhibit discontinuities consistent with the 48 s update interval, while clock corrections show step-like variations every 6 s. Although these fluctuations are at the millimeter-level, they confirm that PPP-B2b corrections provide less smooth continuity than the polynomial-based broadcast ephemerides.

To intuitively demonstrate the effectiveness of PPP-B2b products in reducing asynchronous orbit and clock residual errors, this study evaluates the root mean square (RMS) of radial orbit and clock bias residuals under different data age conditions, using both broadcast ephemeris and PPP-B2b corrections. Considering the variation in error characteristics across different satellite visibility arcs, a segmented arc-based analysis strategy is also applied. As shown in Fig. 8, the asynchronous residuals tend to increase with data age, reflecting the natural degradation of ephemeris accuracy over time. In contrast, the PPP-B2b corrections significantly reduce the residual levels, indicating superior precision performance. However, due to the periodic update nature of PPP-B2b corrections, their short-term continuity is weaker than that of broadcast ephemerides. For example, at a 15 s data age, the PPP-B2b radial orbit residual for G15 and the clock residual for G07 are slightly higher than those of the broadcast ephemeris.

Fig. 8 RMS of asynchronous residuals in radial orbit (top) and clock bias errors (bottom) of broadcast and PPP-B2b ephemeris at different data ages



B2b-ARTK performance under constant data age conditions

Based on the analysis of asynchronous orbit and clock errors at different data ages in Section B, this section evaluates the positioning performance of the Brdc-ARTK and B2b-ARTK models under various constant data age conditions using static reference station data (Dataset I). The evaluation metrics primarily include fixed solution rate and fixed solution accuracy. As shown in Fig. 9, both models exhibit a decreasing trend in fixed rate and fixed solution accuracy as the data age increases. However, compared to the traditional Brdc-ARTK model, the proposed B2b-ARTK model demonstrates significant advantages in both metrics. For example, at a 120-s data age, the fixed rate improves from 79.8 to 96.1%, the horizontal fixed solution accuracy improves from 0.024 m to 0.015 m, and the vertical fixed solution accuracy improves from 0.043 to 0.027 m.

To further investigate short-term performance, Fig. 10 presents the positioning errors under a 3-s constant data age. The results show that when the base and rover use the same correction set, B2b-ARTK achieves almost identical performance to Brdc-ARTK. However, when crossing a correction update boundary, the PPP-B2b solutions exhibit millimeter-level periodic jumps with a cycle of 3 s, reflecting the 6-s clock update interval. Although this limits smoothness under very short data-age conditions, the effect is minor and does not compromise overall positioning accuracy.

Figure 11 further shows that at a 60-s data age, the post-fit carrier-phase residuals of the B2b-ARTK model are significantly smaller than those of the Brdc-ARTK model, indicating better positioning consistency. It should be noted that the PPP-B2b product provides limited short-term accuracy improvement for the CNAV1 ephemeris of the BDS-3 system; its main advantage lies in enhancing the continuity of orbit and clock corrections during ephemeris switching rather than improving absolute accuracy. This is

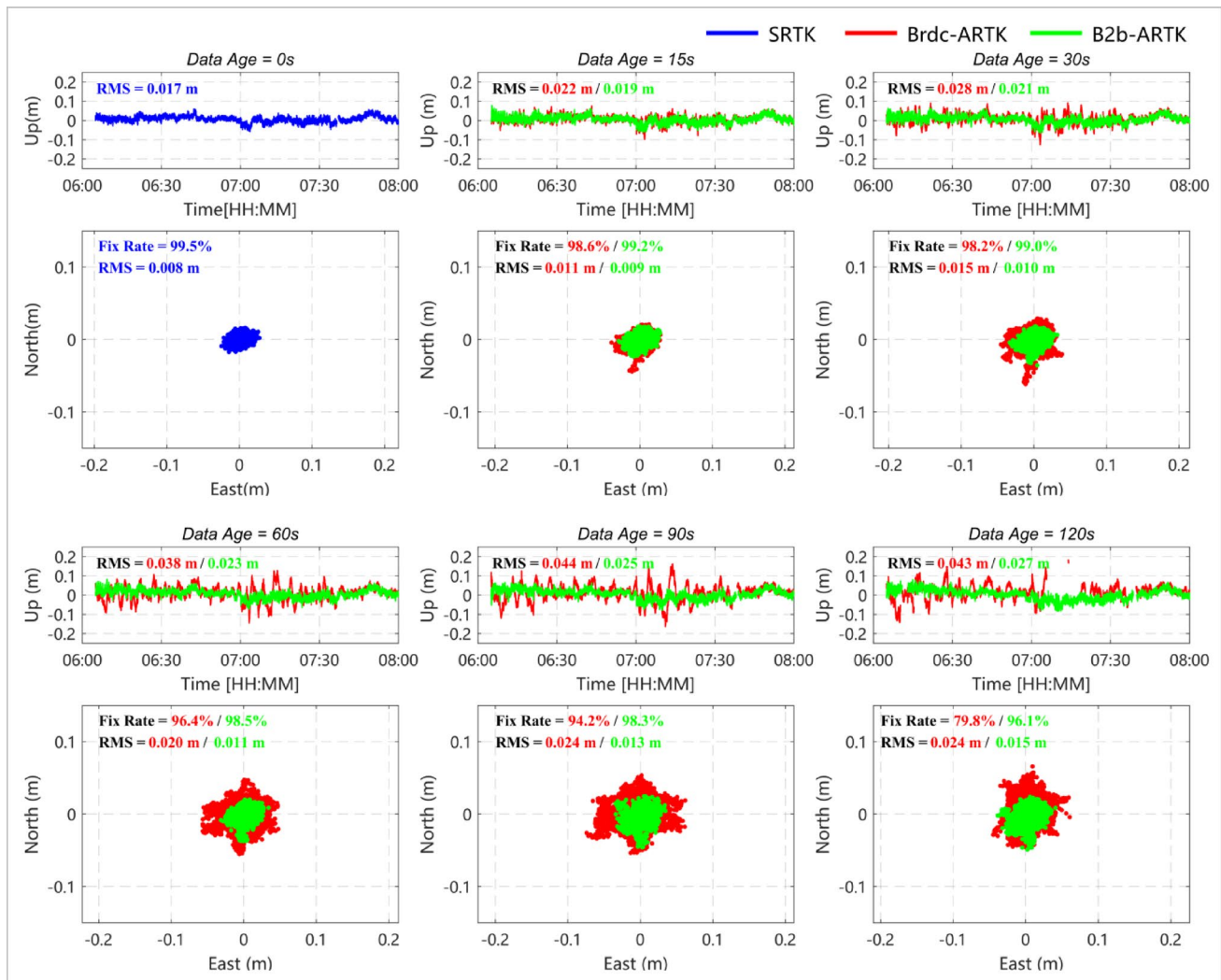


Fig. 9 Positioning performance of Brdc-ARTK and B2b-ARTK models under different constant data age conditions

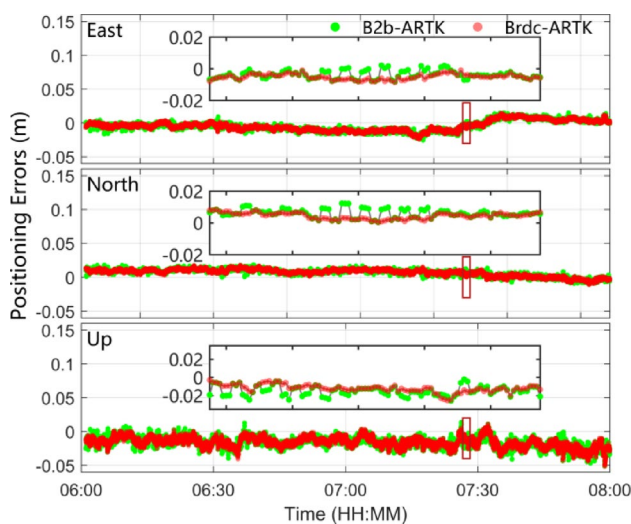


Fig. 10 Positioning errors of Brdc-ARTK and B2b-ARTK models under 3-s constant data age

also reflected in the comparison of post-fit residuals: the improvement for the BDS-3 system is modest (reduced from 0.019 to 0.013 m), whereas the GPS system shows a more pronounced residual reduction (from 0.030 to 0.012 m).

Overall, the experimental results validate that the improvement of broadcast ephemeris orbit and clock accuracy via PPP-B2b effectively reduces residual errors in ARTK and significantly enhances positioning performance. From a practical perspective, broadcast ephemerides can ensure smoother continuity under very low-latency conditions, but PPP-B2b becomes clearly superior as latency increases.

Positioning performance of static scenario under simulated network outage

To further validate the applicability of the proposed model under real network outage scenarios, this study designed an

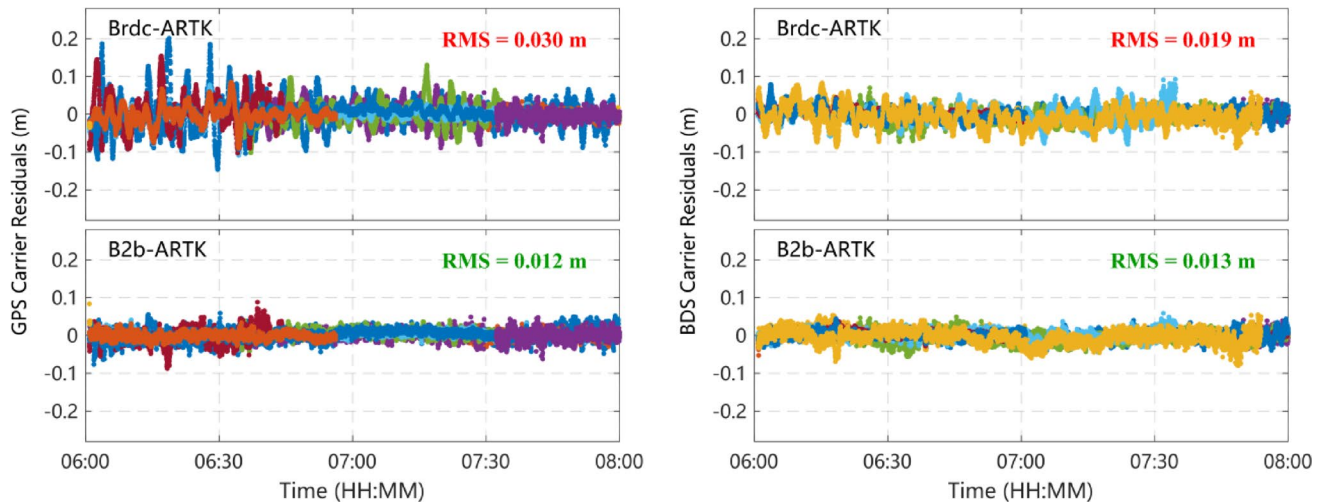


Fig. 11 Comparison of post-fit carrier-phase residuals between B2b-ARTK and Brdc-ARTK models at 60-s data age

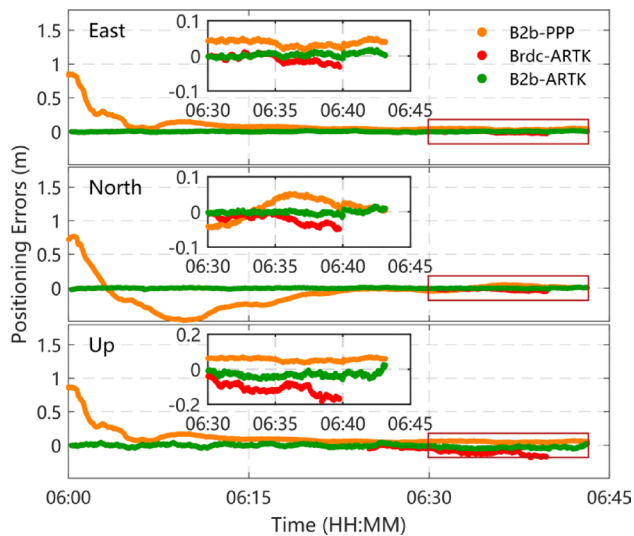


Fig. 12 Positioning error sequences of three models under simulated network outage conditions in the static experiment

Table 4 Positioning accuracy statistics of three models during the asynchronous phase in the static experiment

Models	E/m	N/m	U/m	Max 3D errors/m	Fixed solution duration/s
B2b-PPP	0.0371	0.0306	0.0567	0.0920	—
Brdc-ARTK	0.0138	0.0246	0.1154	0.1899	586
B2b-ARTK	0.0058	0.0047	0.0291	0.0580	790

asynchronous positioning experiment based on static reference station data. The experimental strategy is as follows: during the first 30 min, network connection remains normal and positioning is conducted using the SRTK mode, ensuring full convergence of the B2b-PPP model; from the 30th minute onward, a network outage is simulated, and the Brdc-ARTK and B2b-ARTK models enter the ARTK phase,

with data age increasing continuously over time, while the B2b-PPP model continues continuous positioning. As shown in Fig. 12, after the network outage, the positioning error divergence rate of B2b-ARTK is significantly lower than that of Brdc-ARTK, demonstrating its stronger capability to resist error accumulation.

Table 4 summarizes the positioning accuracy statistics of the three models during the asynchronous phase, as well as the duration for which the two ARTK models maintained fixed solutions. The results show that B2b-ARTK achieved positioning accuracies of 0.006 m (E), 0.005 m (N), and 0.029 m (U), outperforming both Brdc-ARTK (0.014 m, 0.025 m, 0.115 m) and B2b-PPP (0.037 m, 0.031 m, 0.057 m). In terms of error divergence, the maximum 3D positioning error of B2b-ARTK was 0.058 m, which is significantly lower than that of Brdc-ARTK (0.190 m) and B2b-PPP (0.092 m), further demonstrating its stronger ability to suppress error accumulation during the asynchronous phase. Moreover, B2b-ARTK was able to maintain fixed solutions for up to 790 s, compared to only 586 s for Brdc-ARTK.

Furthermore, as illustrated in Fig. 13, the carrier-phase post-fit residuals of the Brdc-ARTK model exhibit a continuously diverging trend with increasing data age (indicated by the yellow shaded area), reaching approximately 0.2 m after 30 min. In contrast, the B2b-ARTK model effectively suppresses the accumulation of residuals, maintaining them within 0.1 m throughout the period. This further confirms the significant contribution of PPP-B2b corrections to enhancing the positioning performance of ARTK.

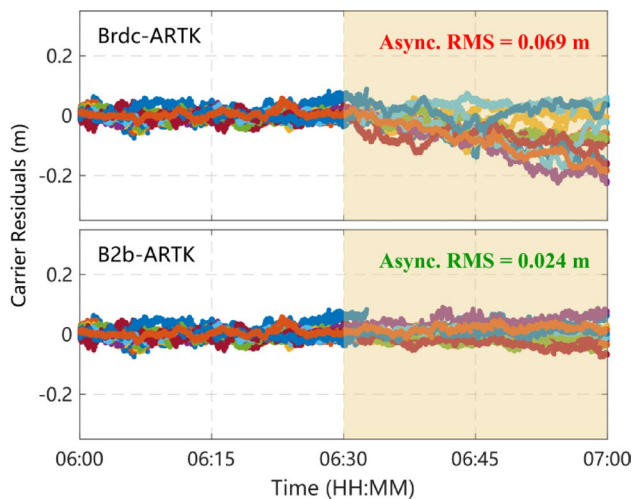


Fig. 13 Post-fit carrier-phase residuals of the Brdc-ARTK and B2b-ARTK models under simulated network outage conditions in the static experiment

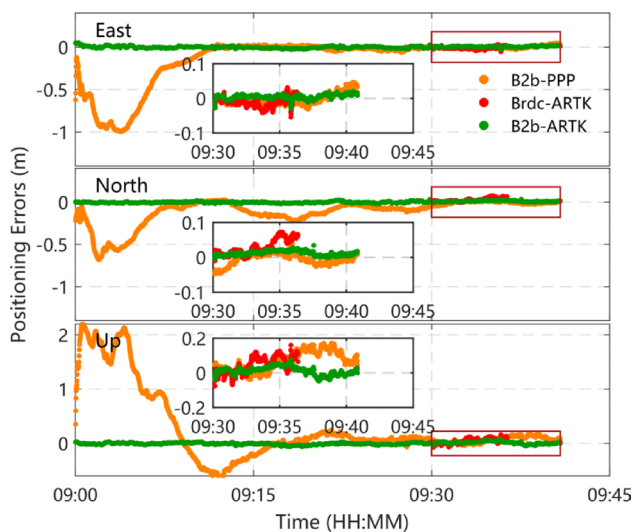


Fig. 14 Positioning error sequences of three models under simulated network outage conditions in the dynamic experiment

Positioning performance of dynamic scenario under simulated network outage

To further validate the positioning performance of the proposed model under dynamic network outage conditions, ARTK experiments were conducted using a set of dynamic vehicle data. The experimental setup followed the same strategy as in Section D. Figure 14 illustrates the time series of the three-dimensional positioning errors for the three models during the asynchronous phase. It can be observed that, throughout the dynamic driving process, the positioning errors of the Brdc-ARTK model rapidly diverge as the data age increases, whereas the B2b-ARTK model effectively

Table 5 Positioning accuracy statistics of three models during the asynchronous phase in the dynamic experiment

Models	E/m	N/m	U/m	Max 3D errors/m	Fixed solution duration/s
B2b-PPP	0.0257	0.0271	0.0974	0.1725	–
Brdc-ARTK	0.0283	0.0350	0.0758	0.1777	382
B2b-ARTK	0.0182	0.0233	0.0353	0.0847	650

suppresses error accumulation over the entire asynchronous period, demonstrating superior dynamic stability.

Table 5 summarizes the positioning accuracy and duration of fixed solution maintenance for the three models during the asynchronous phase. The B2b-ARTK model achieved average positioning errors of 0.018 m, 0.023 m, and 0.035 m in the east, north, and up directions, respectively, with a maximum 3D error of 0.085 m, outperforming the Brdc-ARTK model (0.028 m, 0.035 m, 0.076 m; 0.178 m) and the B2b-PPP model (0.026 m, 0.027 m, 0.097 m; 0.173 m). Furthermore, the B2b-ARTK model maintained a fixed solution for up to 650 s, significantly longer than the 382 s sustained by the Brdc-ARTK model, further demonstrating its superior capability in maintaining high-precision positioning in dynamic asynchronous environments.

Discussion

To further verify the applicability and generality of the proposed ARTK framework, additional experiments were carried out using both PPP-B2b and Galileo HAS corrections. Two representative test baselines were considered. The first was the HKWS–HKSS baseline in Hong Kong, with a length of approximately 5.6 km, representing the Asia–Pacific region where PPP-B2b provides guaranteed service. The second was the GOPE–GOP6 baseline in Europe, with a length below 10 m, representing the service region of Galileo HAS.

Asia–Pacific comparison

For the HKWS–HKSS baseline, two sets of solutions were compared: PPP-B2b corrections against GPS + BDS broadcast ephemerides, and HAS corrections against GPS + Galileo broadcast ephemerides. The experiments were conducted under a 60 s constant-age positioning mode. As shown in Fig. 15, PPP-B2b corrections significantly reduced positioning errors compared with broadcast products, while HAS corrections in the Asia–Pacific region showed only limited improvement. This is consistent with the Galileo HAS service specification, which does not guarantee correction accuracy in this region.

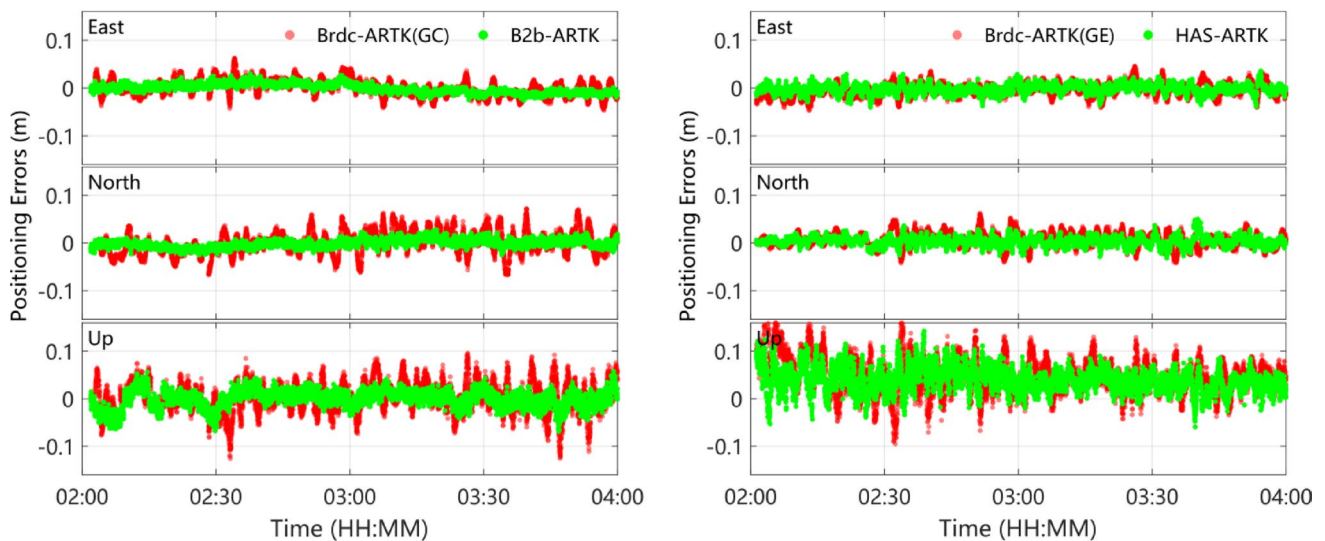


Fig. 15 Positioning errors under 60-s constant-age conditions for the HKWS–HKSS baseline in the Asia–Pacific region: (left) PPP-B2b corrections versus GPS+BDS broadcast ephemerides; (right) Galileo HAS corrections versus GPS+Galileo broadcast ephemerides

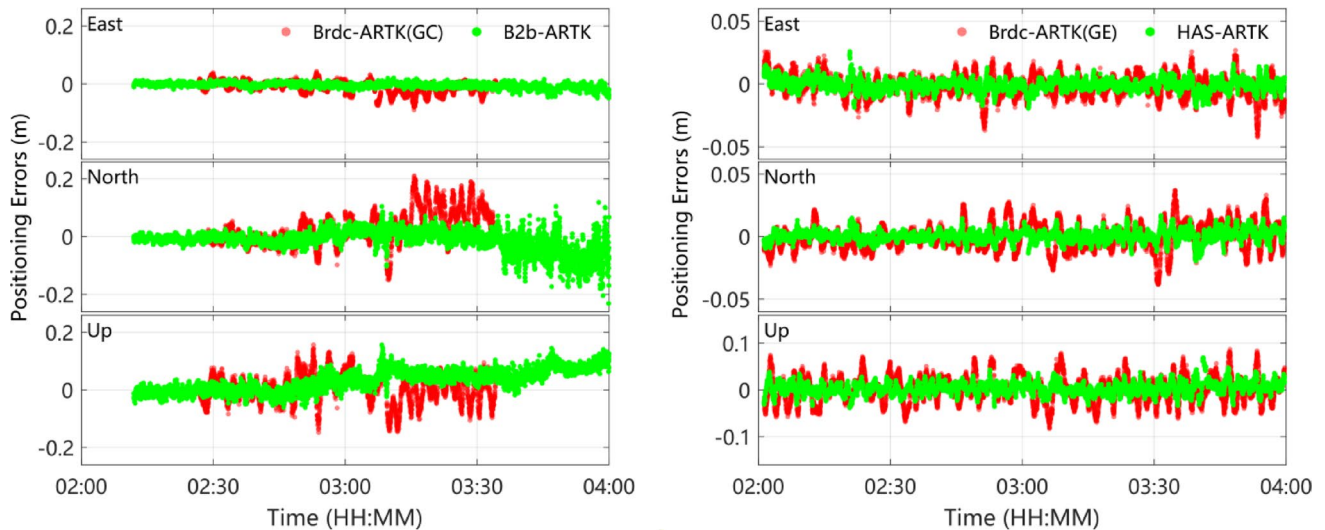


Fig. 16 Positioning errors under 60-s constant-age conditions for the GOPE–GOP6 baseline in Europe: (left) PPP-B2b corrections versus GPS+BDS broadcast ephemerides; (right) Galileo HAS corrections versus GPS+Galileo broadcast ephemerides

European comparison.

For the GOPE–GOP6 short baseline in Europe, Fig. 16 shows that Galileo HAS corrections demonstrated substantial improvement relative to GPS+Galileo broadcast ephemerides, achieving centimeter- to millimeter-level positioning accuracy (Fig. 16). This result validates the effectiveness of HAS in its guaranteed service area. In contrast, the performance of PPP-B2b in Europe was constrained by the limited availability of BDS satellites, which led to longer convergence times and reduced positioning stability. This reflects the regional complementarity of the two services: PPP-B2b is highly advantageous in the Asia–Pacific, while Galileo HAS provides superior corrections in

Europe, and both can be seamlessly integrated within the proposed ARTK framework.

Table 6 summarizes the positioning accuracy, ambiguity-fixing rate, and number of available satellites (AVS) for all tested cases. These results clearly show that:

1. PPP-B2b corrections are highly effective in the Asia–Pacific region, outperforming both broadcast ephemerides and HAS.
2. HAS corrections demonstrate significant improvements in Europe, where their service performance is guaranteed.

Table 6 Summary of positioning performance in Asia–Pacific (HKWS–HKSS) and Europe (GOPE–GOP6) under different correction strategies, including positioning accuracy, ambiguity-fixing rate, and number of available satellites (AVS)

Baseline	Model	Positioning errors/m			Fix Rate	AVS
		E	N	U		
HKWS–HKSS (5.6 km)	Brdc-ARTK(GC)	0.016	0.023	0.033	97.45%	13.7
	B2b-ARTK	0.009	0.008	0.016	98.94%	
	Brdc-ARTK(GE)	0.014	0.016	0.063	87.46%	
	HAS-ARTK	0.010	0.016	0.047	91.02%	
GOPE–GOP6 (<10 m)	Brdc-ARTK(GC)	0.023	0.065	0.056	52.94%	7.7
	B2b-ARTK	0.010	0.038	0.049	89.92%	
	Brdc-ARTK(GE)	0.009	0.010	0.284	97.54%	
	HAS-ARTK	0.004	0.004	0.013	98.67%	

3. The proposed ARTK framework works consistently with both correction sources, confirming its generality across different satellite-based PPP services.

Summary and conclusions

This study proposes an ARTK framework enhanced by satellite-broadcast PPP corrections and implements it with BDS-3 PPP-B2b as a representative case. The framework is motivated by a principled analysis of asynchronous residual formation: when reference and rover epochs are not synchronized, double-differencing cannot completely cancel orbit, clock, ionospheric, and tropospheric terms. The proposed method therefore introduces explicit mitigation for these four classes of errors and integrates space-based corrections that are independent of terrestrial networks, so that deployment does not require any modification to existing NRTK links. Although PPP-B2b is used for instantiation, the algorithmic design is general and directly extensible to other satellite-based PPP services such as Galileo HAS.

The effectiveness of the framework is substantiated by a coherent suite of tests that bridge product-level accuracy and positioning-level performance. First, orbit and clock evaluations show that PPP-B2b markedly reduces the standard deviations of radial orbit and clock errors for both GPS and BDS-3, while revealing short-term update behavior that is relevant to ARTK. Second, constant-age positioning experiments quantify availability and accuracy across latency regimes; at 120 s data age the fixing rate increases from 79.8 to 96.1%, and horizontal and vertical accuracies improve from 0.024 and 0.043 m to 0.015 m and 0.027 m. Third, simulated network-outage scenarios in static and dynamic settings demonstrate that the proposed B2b-ARTK sustains fixed solutions for 790 s and 650 s, with maximum 3D errors of 0.058 m and 0.085 m, respectively, thereby evidencing robustness under communication instability. Complementary analyses further indicate that the same framework applies to other satellite-broadcast PPP services within their guaranteed service regions.

In summary, the proposed satellite-based PPP-enhanced ARTK effectively mitigates accuracy degradation at high latency and is well suited to real deployments in autonomous driving, precision mapping, and engineering monitoring. Future work will incorporate explicit modeling of correction uncertainty and short-term prediction to handle rare interruptions of the space-based correction broadcast itself (Xu et al. 2024).

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Author contribution HH and WG wrote the main manuscript text. SP and LQ prepared Figs. 1, 2, 3, 4, 5, 6, 7, 8 and 9. QZ prepared Figs. 10, 11, 12, 13, 14, 15 and 16. All authors reviewed the manuscript.

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Data availability The data that support the findings of this study are available from the corresponding author upon request.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

Consent for publication All authors have read and agreed to the published version of the manuscript.

Ethical approval Not applicable.

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